MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3690 GOLD ROOM

DESIGN CALCULATION

AS BUILT

MD-216-3690-EG-CV-CAL-1372

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2		
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 306 :2007	Saudi Steel Structures Code
AISC LRFD	Building Code Requirement of Steel Structural Building
AISC Steel Design Guide 1 - 2nd Edition	Base Plate and Anchor Rod Design

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for design of Gold Room Stair Block Foundations & Pedestals

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The document is prepared for design of Gold Room Stair Block Foundations & Pedestals
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
3D- Model and analysis have been performed in Staad-Pro Software.
Foundation & Pedestals have been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.
Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4 DESIGN OF RCC PORTION FOR STAIR BLOCK

4.1 Design — Design of RCC Str. and its components shall be designed in accordance with

[1] SBC-301-2007-Structural Loading & Forces.
[2] SBC-303-2007-Structural Soil & Foundations.
[3] SBC-304-2018-Structural Concrete Structures.
[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.
[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete
[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.
[7] ACI 318-08 Building Code Requirements.
[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

4.2 Materials

Compressive Strength of Concrete at 28th days (fc') = 35 MPa
Yield Strength of Steel (fy) = 420 MPa

4.3 Reference — Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 SUMMARY OF SUPPORT REACTIONS **(Refer Staad File - GOLD ROOM STAIR BLOCK R0.STD)**

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK : **(At Bottom of Base Plate)**

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	3	128 0.6 (DL +	**4.596**	14.687	-0.117	0	0	0
Min Fx	3	123 0.6 (DL +	**-4.593**	-6.321	-0.134	0	0	0
Max Fy	4	107 DL + SDL	-0.063	**18.679**	-0.348	0	0	0
Min Fy	4	128 0.6 (DL +	0.076	**-6.829**	0.228	0	0	0
Max Fz	4	130 0.6 (DL +	-0.002	1.416	**1.823**	0	0	0
Min Fz	4	125 0.6 (DL +	-0.003	6.645	**-1.865**	0	0	0
Max Mx	1	101 DL+SDL1	-0.043	2.807	0.123	**0**	0	0
Min Mx	1	101 DL+SDL1	-0.043	2.807	0.123	**0**	0	0
Max My	1	101 DL+SDL1	-0.043	2.807	0.123	0	**0**	0
Min My	1	101 DL+SDL1	-0.043	2.807	0.123	0	**0**	0
Max Mz	1	101 DL+SDL1	-0.043	2.807	0.123	0	0	**0**
Min Mz	1	101 DL+SDL1	-0.043	2.807	0.123	0	0	**0**

Pedestal Height = 2 m

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK : **(At Foundaing Level)**

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	21	132 0.6 (DL +	**4.596**	14.687	-0.117	-0.234	0.000	9.192
Min Fx	27	135 0.6 (DL +	**-4.593**	-6.321	-0.134	-0.268	0.000	-9.186
Max Fy	22	103 DL + SDL	-0.063	**18.679**	-0.348	-0.696	0.000	-0.126
Min Fy	31	118 DL + SDL	0.076	**-6.829**	0.228	0.456	0.000	0.152
Max Fz	27	122 DL + SDL	-0.002	1.416	**1.823**	3.646	0.000	-0.004
Min Fz	20	117 DL + SDL	-0.003	6.645	**-1.865**	-3.730	0.000	-0.006
Max Mx	27	138 0.6 (DL +	-0.043	2.807	0.123	**0.246**	0.000	-0.086
Min Mx	192	117 DL + SDL	-0.043	2.807	0.123	**0.246**	0.000	-0.086
Max My	30	131 0.6 (DL +	-0.043	2.807	0.123	0.246	**0.000**	-0.086
Min My	28	132 0.6 (DL +	-0.043	2.807	0.123	0.246	**0.000**	-0.086
Max Mz	21	135 0.6 (DL +	-0.043	2.807	0.123	0.246	0.000	**-0.086**
Min Mz	21	132 0.6 (DL +	-0.043	2.807	0.123	0.246	0.000	**-0.086**

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SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN : **(At Bottom of Base Plate)**

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	3	212 1.2 (DL +	**7.446**	27.409	-0.22	0	0	0
Min Fx	3	207 1.2 (DL +	**-7.433**	-6.717	-0.291	0	0	0
Max Fy	4	207 1.2 (DL +	-0.13	**27.77**	-0.495	0	0	0
Min Fy	4	228 0.9 (DL +	0.122	**-11.229**	0.353	0	0	0
Max Fz	4	214 1.2 (DL +	-0.004	5.414	**2.966**	0	0	0
Min Fz	4	209 1.2 (DL +	-0.006	14.578	**-3.046**	0	0	0
Max Mx	1	201 1.4 (DL+S	-0.061	3.93	0.172	**0**	0	0
Min Mx	1	201 1.4 (DL+S	-0.061	3.93	0.172	**0**	0	0
Max My	1	201 1.4 (DL+S	-0.061	3.93	0.172	0	**0**	0
Min My	1	201 1.4 (DL+S	-0.061	3.93	0.172	0	**0**	0
Max Mz	1	201 1.4 (DL+S	-0.061	3.93	0.172	0	0	**0**
Min Mz	1	201 1.4 (DL+S	-0.061	3.93	0.172	0	0	**0**

Pedestal Height = 2 m

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN : **(At Foundaing Level)**

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	3	212 1.2 (DL +	**7.446**	27.409	-0.22	-0.440	0.000	14.892
Min Fx	3	207 1.2 (DL +	**-7.433**	-6.717	-0.291	-0.582	1.000	-14.866
Max Fy	4	207 1.2 (DL +	-0.13	**27.77**	-0.495	-0.990	2.000	-0.260
Min Fy	4	228 0.9 (DL +	0.122	**-11.229**	0.353	0.706	3.000	0.244
Max Fz	4	214 1.2 (DL +	-0.004	5.414	**2.966**	5.932	4.000	-0.008
Min Fz	4	209 1.2 (DL +	-0.006	14.578	**-3.046**	-6.092	5.000	-0.012
Max Mx	1	201 1.4 (DL+S	-0.061	3.93	0.172	**0.344**	6.000	-0.122
Min Mx	1	201 1.4 (DL+S	-0.061	3.93	0.172	**0.344**	7.000	-0.122
Max My	1	201 1.4 (DL+S	-0.061	3.93	0.172	0.344	**8.000**	-0.122
Min My	1	201 1.4 (DL+S	-0.061	3.93	0.172	0.344	**9.000**	-0.122
Max Mz	1	201 1.4 (DL+S	-0.061	3.93	0.172	0.344	10.000	**-0.122**
Min Mz	1	201 1.4 (DL+S	-0.061	3.93	0.172	0.344	11.000	**-0.122**

6 DESIGN OF RCC ELEMENT

Foundation & Pedestals have been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.

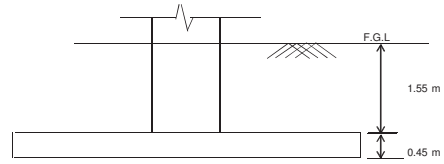
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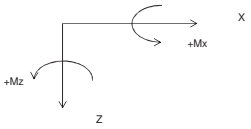
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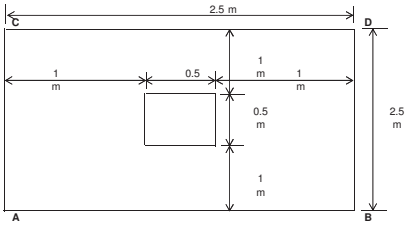
6.1 DESIGN OF FOUNDATION

MKD. F4A (NODE NO. : 1 to 4)

Net Safe Bearing Capacity of Soil =	40	Mton/m^2	
Depth of Foundation from FGL =	2	m	
Gross Allowable Bearing Capacity of Soil =	40 + 2 X 1.9 =	43.8	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	2.5	m
Width of Foundation (Lz) =	2.5	m
Thickness of Base Raft =	0.45	m
Length of Column (lx) =	0.5	m
Width of Column (lz) =	0.5	m
Area , A = Lx*Lz		
=	6.25	m^2
Zxx = Lx*Lz^2/6		
=	2.604	m^3
Zzz = Lz*Lx^2/6		
=	2.604	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2.5X2.5X0.45X2.5 =	7.031
Column =	0.5X0.5X1.55X2.5 =	0.969
Soil =	((2.5X2.5)-(0.5X0.5))X1.55X1.9 =	17.67
	Total =	25.67

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	21	132 0.6 (DL +	**4.596**	14.687	-0.117	-0.234	0.000	9.192
Min Fx	27	135 0.6 (DL +	**-4.593**	-6.321	-0.134	-0.268	0.000	-9.186
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Max Fz	27	122 DL + SDL	-0.002	1.416	**1.823**	3.646	0.000	-0.004
Min Fz	20	117 DL + SDL	-0.003	6.645	**-1.865**	-3.730	0.000	-0.006
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Max My	30	131 0.6 (DL +	-0.043	2.807	0.123	0.246	**0.000**	-0.086
Min My	28	132 0.6 (DL +	-0.043	2.807	0.123	0.246	**0.000**	-0.086
Max Mz	21	135 0.6 (DL +	-0.043	2.807	0.123	0.246	0.000	**-0.086**
Min Mz	21	132 0.6 (DL +	-0.043	2.807	0.123	0.246	0.000	**-0.086**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
132	40.357	4.597	0.5	20.179	4.39	1.5	**O.K.**
135	19.349	4.595	0.5	9.675	2.11	1.5	**O.K.**
103	44.349	0.354	0.5	22.175	62.64	1.5	**O.K.**
118	18.841	0.24	0.5	9.421	39.25	1.5	**O.K.**
122	27.086	1.823	0.5	13.543	7.43	1.5	**O.K.**
117	32.315	1.865	0.5	16.158	8.66	1.5	**O.K.**
138	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**
117	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**
131	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**
132	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**
135	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**
132	28.477	0.13	0.5	14.239	109.53	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 118 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)

Max Upword Force =	6.829	Mton			
Self Wt. of Foundation =	25.67	Mton			
Overturning Moment =	6.829 X 2.5/2 + 0.456 =		8.992	Mton-m	
Resisting Moment =	25.67 X 0.9 X 2.5/2 =		28.879	Mton-m	
F.O.S =	3.21	>.	2	**O.K.**	

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zxx (Mton/m³)	Mz/Zzz (Mton/m³)	BASE PRESSURE AT POINT A	B	C	D	Remarks
132	6.457	0.09	3.53	10.077	3.017	9.897	2.837	**O.K.**
135	3.096	0.103	3.528	6.727	**-0.329**	6.521	**-0.535**	**O.K.**
103	7.096	0.267	0.048	7.411	7.315	6.877	6.781	**O.K.**
118	3.015	0.175	0.058	3.248	3.132	2.898	2.782	**O.K.**
122	4.334	1.4	0.002	5.736	5.732	2.936	2.932	**O.K.**
117	5.17	1.432	0.002	6.604	6.6	3.74	3.736	**O.K.**
138	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
117	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
131	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
132	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
135	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
132	4.556	0.094	0.033	4.683	4.617	4.495	4.429	**O.K.**
			MAX =	10.077	7.315	9.897	6.781	

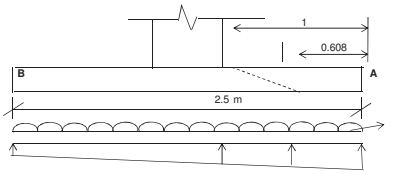
SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	21	SDL1) + 0.5 LL	**7.446**	27.409	-0.22	-0.44	0	14.892
Min Fx	27	DL + SDL1) + 1	**-7.433**	-6.717	-0.291	-0.582	1	-14.866
Max Fy	22	L + SDL1 + TL	-0.13	**27.77**	-0.495	-0.99	2	-0.26
Min Fy	31	L + SDL1 + TL	0.122	**-11.229**	0.353	0.706	3	0.244
Max Fz	27	SDL2) + 0.5 LL	-0.004	5.414	**2.966**	5.932	4	-0.008
Min Fz	20	SDL1) + 0.5 LL	-0.006	14.578	**-3.046**	-6.092	5	-0.012
Max Mx	27	SDL2) + 0.5 LL	-0.061	3.93	0.172	**0.344**	6	-0.122
Min Mx	192	SDL1) + 0.5 LL	-0.061	3.93	0.172	**0.344**	7	-0.122
Max My	30	DL + SDL1) + 1	-0.061	3.93	0.172	0.344	**8**	-0.122
Min My	28	SDL1) + 0.5 LL	-0.061	3.93	0.172	0.344	**9**	-0.122
Max Mz	21	DL + SDL1) + 1	-0.061	3.93	0.172	0.344	10	**-0.122**
Min Mz	21	SDL1) + 0.5 LL	-0.061	3.93	0.172	0.344	11	**-0.122**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zx (Mton/m³)	Mz/Zz (Mton/m³)	BASE PRESSURE AT POINT A	B	C	D
216	8.493	0.169	5.719	14.381	2.943	14.043	2.605
231	3.032	0.224	5.709	8.965	-2.453	8.517	-2.901
203	8.55	0.38	0.1	9.03	8.83	8.27	8.07
203	2.311	0.271	0.094	2.676	2.488	2.134	1.946
222	4.973	2.278	0.003	7.254	7.248	2.698	2.692
217	6.44	2.339	0.005	8.784	8.774	4.106	4.096
222	4.736	0.132	0.047	4.915	4.821	4.651	4.557
217	4.736	0.132	0.047	4.915	4.821	4.651	4.557
231	4.736	0.132	0.047	4.915	4.821	4.651	4.557
216	4.736	0.132	0.047	4.915	4.821	4.651	4.557
231	4.736	0.132	0.047	4.915	4.821	4.651	4.557
216	4.736	0.132	0.047	4.915	4.821	4.651	4.557

PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point B (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	14.381	2.943	9.81	11.6	6.679	2.248
231	8.965	-2.453	4.4	6.19	3.969	1.247
203	9.03	8.83	8.95	8.98	2.507	0.923
203	2.676	2.488	2.6	2.63	-0.672	-0.252
222	7.254	7.248	7.25	7.25	1.593	0.589
217	8.784	8.774	8.78	8.78	2.358	0.872
222	4.915	4.821	4.88	4.89	0.434	0.159
217	4.915	4.821	4.88	4.89	0.434	0.159
231	4.915	4.821	4.88	4.89	0.434	0.159
216	4.915	4.821	4.88	4.89	0.434	0.159
231	4.915	4.821	4.88	4.89	0.434	0.159
216	4.915	4.821	4.88	4.89	0.434	0.159
				MAX =	6.679	2.248

ALONG X - DIRECTION :

Mu = 6.679 Mton-m

ΦᵏKn = Mu/bd² deff. = 392 mm

Mu/bd² = 6.679X10^7/(1000X392^2) = 0.435 N/mm² < 6.9 N/mm²

Pt = 0.0012

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X392/100 = 7.06 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0034

Vu = 2.248 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0034X2.248X10^4X392/6.679/10^7)X1000X392/10^4
= 37.405 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4
= 39.371 Mton

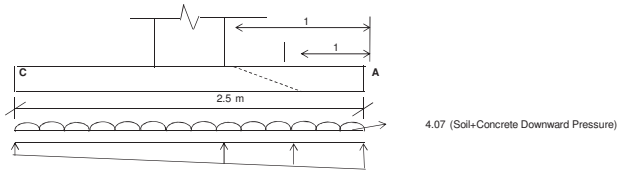
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X392/10^4
= 67.254 Mton

HENCE, Vc = 37.405 Mton

Φt*Vc = 0.75X37.405 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 28.054 Mton **> 2.248Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.405X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 117.601 Mton **> Vu= 2.248Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point C (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	14.381	14.043	14.25	14.25	5.199	4.918
231	8.965	8.517	8.79	8.79	2.506	2.133
203	9.03	8.27	8.73	8.73	2.58	1.947
203	2.676	2.134	2.46	2.46	-0.625	-1.077
222	7.254	2.698	5.43	5.43	2.2	-1.597
217	8.784	4.106	6.91	6.91	2.982	-0.917
222	4.915	4.651	4.81	4.81	0.458	0.238
217	4.915	4.651	4.81	4.81	0.458	0.238
231	4.915	4.651	4.81	4.81	0.458	0.238
216	4.915	4.651	4.81	4.81	0.458	0.238
231	4.915	4.651	4.81	4.81	0.458	0.238
216	4.915	4.651	4.81	4.81	0.458	0.238
				MAX =	5.199	4.918

ALONG Z - DIRECTION :

Mu = 5.199 T-m

Φ*Kn = Mu/bd² deff. = 392 mm

Mu/bd² = 5.199X10^7/(1000X392^2) = 0.338 N/mm² < 6.9 N/mm²

Pt = 0.0009

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X392/100 = 7.06 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 4.918 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0034X4.918X10^4X392/5.199/10^7)X1000X392/10^4
= 37.946 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4
= 39.371 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X392/10^4
= 67.254 Mton

HENCE, Vc = 37.946 Mton

Φt*Vc = 0.75X37.946 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 28.46 Mton **> 4.918Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.946X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 117.642 Mton **> Vu= 4.918Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 6.25 m²

Perimeter (bo) = 2 X ((0.5 + 0.392) + (0.5 + 0.392)) = 3.568 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 10.077 Mton/m²

Vu = ((6.25 - (0.5X0.5))X10.077
= 60.462 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X3.568X10^3X392/10^4
= 273.06 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1)X1X35^0.5X3.568X10^3X392/10^4
= 422.003 Mton

[c] = 0.083*(2+αs*d.eff/bo)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+30X392/(3.568X10^3))X1X35^0.5X(3.568X10^3)X392/10^4
= 363.721 Mton

HENCE, Vc = 273.06 Mton

Φt*Vn = Φt*Vc = 0.75X273.06 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
204.795 Mton **> 60.462Mton , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	30-05-2020	29-06-2020	28.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT
Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI

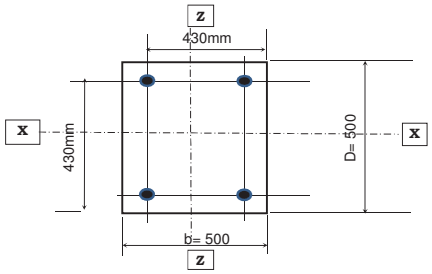
Outotec / L&T Revision	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT
Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI

6.2 DESIGN OF PEDESTAL (Bridge/Drive in Outer Column)

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	70mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor *f* = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) = 2000mm
Unsupproted length of column about Z (l_{uz}) = 2000mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio,	$K l_{ux} / r_x$ =	27.72	< 22 SECTION 6.2.5a	Radius of gyration r_x=	144	mm
	$K l_{uz} / r_z$ =	27.72	< 22 SECTION 6.2.5a	Radius of gyration r_z=	144	mm

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = $Pn/f'_c bD$ (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
212	1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)	28.00	12.0	1.0	43.1	18.5	1.5	0.05

Reinforcement: Provide,

4-Nos	20 mm Dia
2-Nos	20 mm Dia
2-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
628.57 Sq mm (Along Z axis)
628.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2514.29 Sq mm
Total A_{sc} = 1.0 %

b = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 841.87 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	b	$a_n = \log(0.5)/\log(b)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
212	0.10	0.10	43.8	43.8	0.55	1.16	0.39	O.K

Links, 10mm dia @ 300 C/C

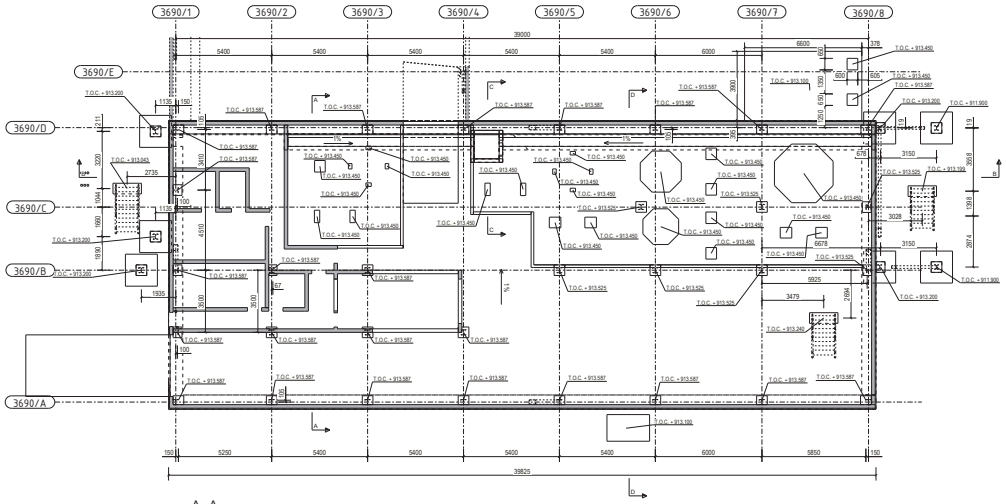
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

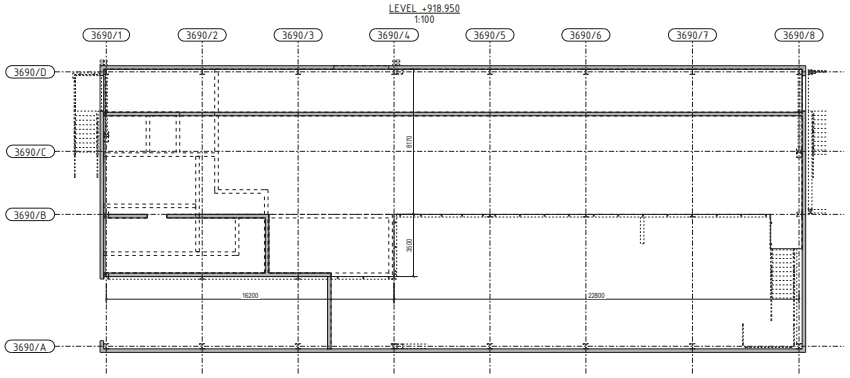
ATTACHMENT 1 : MECHANICAL GENERAL ARRANGEMENT DRAWING

DRAWING NUMBER: MD-216-3690-EG-CV-GAD-1001 A01

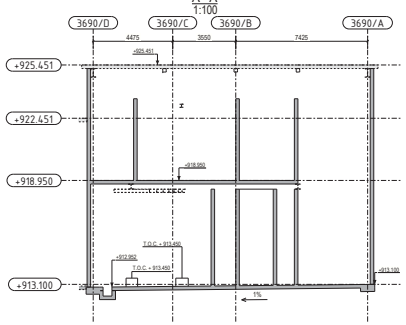
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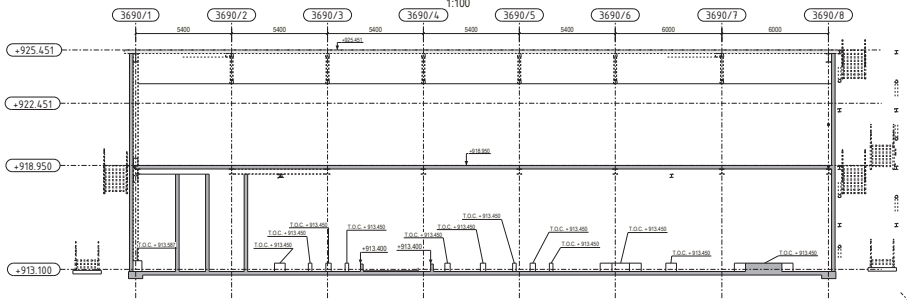
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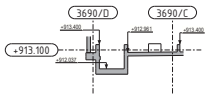
A - A
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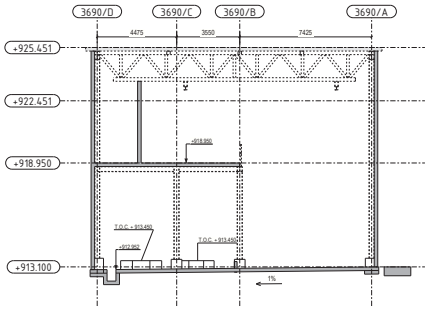
B - B
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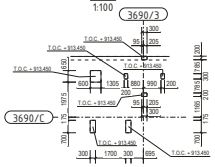
C - C
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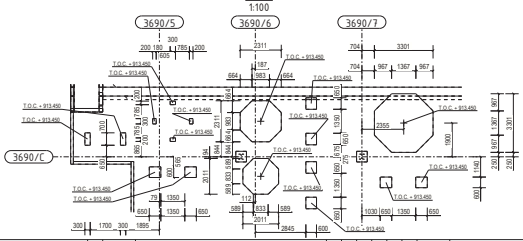
D - D
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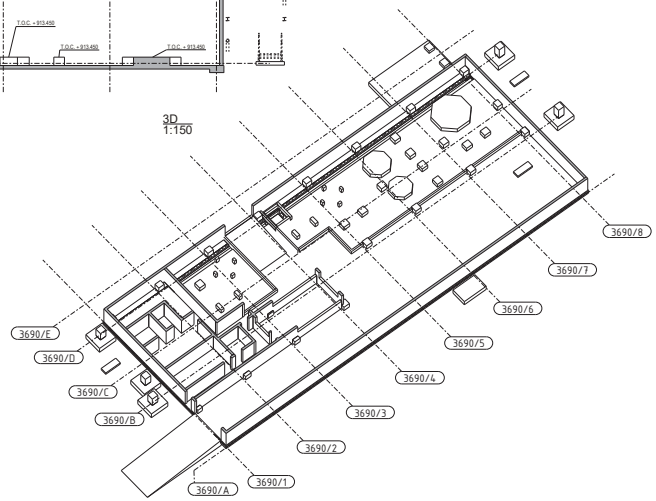
DET 1
1:100



DET 2
1:100



3D
1:150



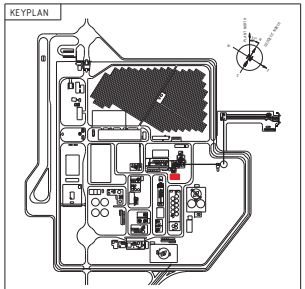
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3690 = RL(+)913.000 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.251, SECTIONS A-A, B-B AND C-C	MD-216-3690-EG-ST-GAD-1001	PLAN AT EL +925.251, SECTIONS A-A - E-E
MD-216-3690-EG-CV-GAD-1002	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1001	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1003	SECTION B-B	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A
MD-216-9000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1006	PLAN AT EL +918.950 & +925.450

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	02.04.2020	ISSUED FOR APPROVAL	Ravish	Mehul	Sudha	Sandip	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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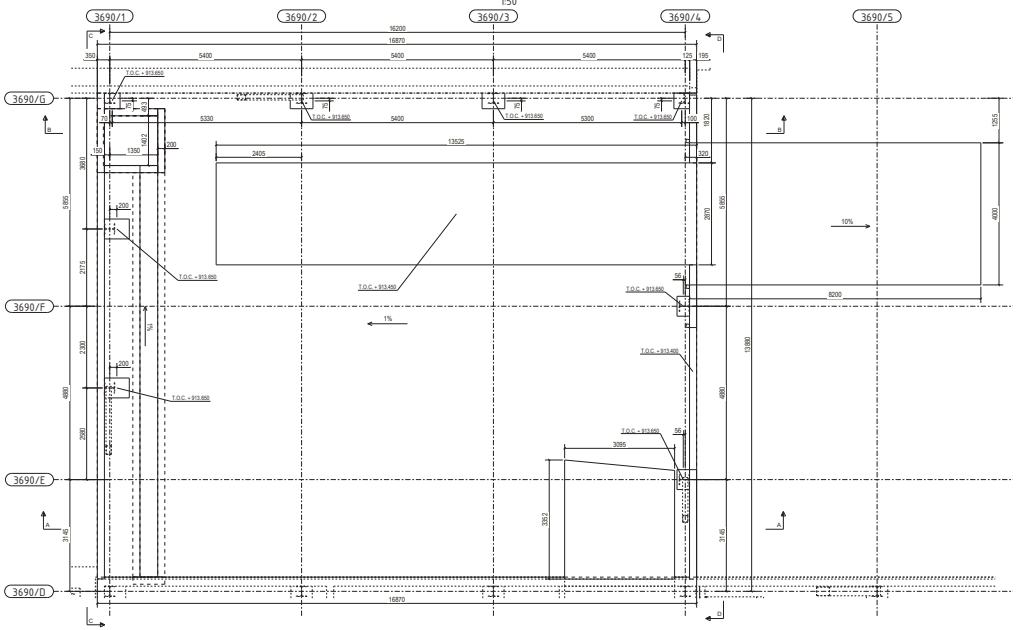
LARSEN & TOUBRO DRAWING NUMBER: 900707_GAU01_EE14_CTA04_36901 — REVISION: 0
OUTOTEC DRAWING NUMBER: OU100668230 — DOCUMENT ID — REVISION: 0

DRAWN	Ajit MK	02.04.2020
CHECKED	Mehul	02.04.2020
ENGINEER	Sudha	02.04.2020
APPROVED	Sandip	02.04.2020
EMPLOYER		
SCALE	1:100	A0

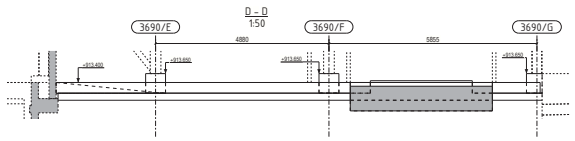
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +913.100

DRAWING NUMBER: MD-216-3690-EG-CV-GAD-1001 — REV: A01

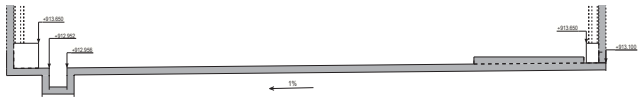
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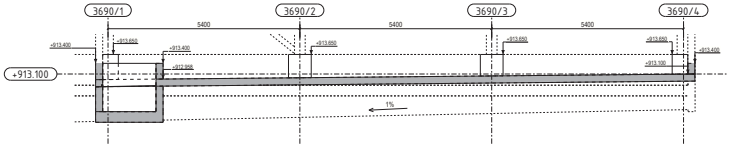
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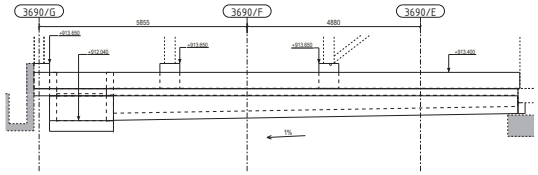
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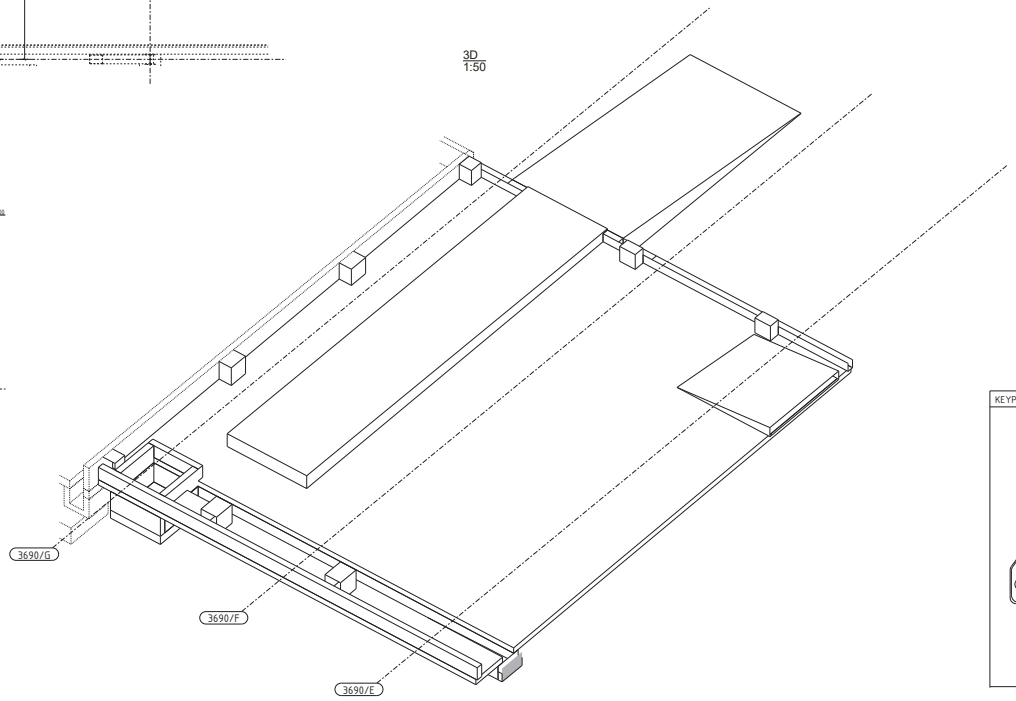
B – B
1:50



C – C
1:50



3D
1:50



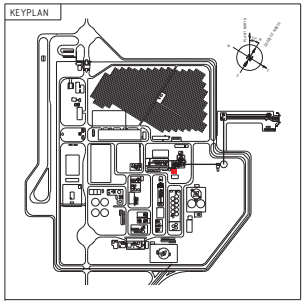
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3620 = RL(+)913.100 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.051, SECTIONS A-A, B-B AND C-C				
MD-216-3690-EG-CV-GAD-1001	PLAN AT EL +913.100	MD-216-3690-EG-ST-GAD-1001	PLAN AT EL +905.201, SECTIONS A-A , E-E		
MD-216-3690-EG-ME-GAD-1001	SECTION B-B	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL +913.100		
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A		
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +918.850 & +915.450		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01 02.04.2020 ISSUED FOR APPROVAL	Ershin	Mekai	Lutfu	Soulia	

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER
90D707_GAU01_EE14_CTA04_36902
REVISION 0

OUTOTEC DRAWING NUMBER
OU100668231
DOCUMENT ID
REVISION 0

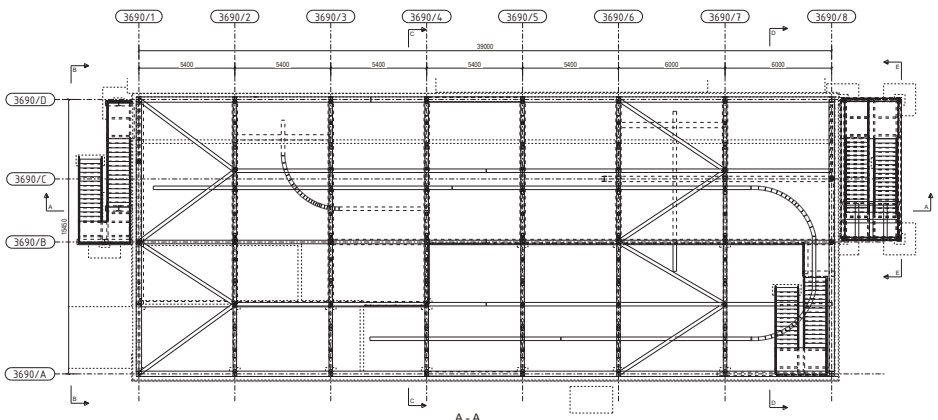
DRAWN	Ad Ak	02.04.2020
CHECKED	Mekai	02.04.2020
ENGINEER	Lutfu	02.04.2020
APPROVED	Soulia	02.04.2020
EMPLOYER		
SCALE	1:50	A0

MANSOURAH – MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +913.100

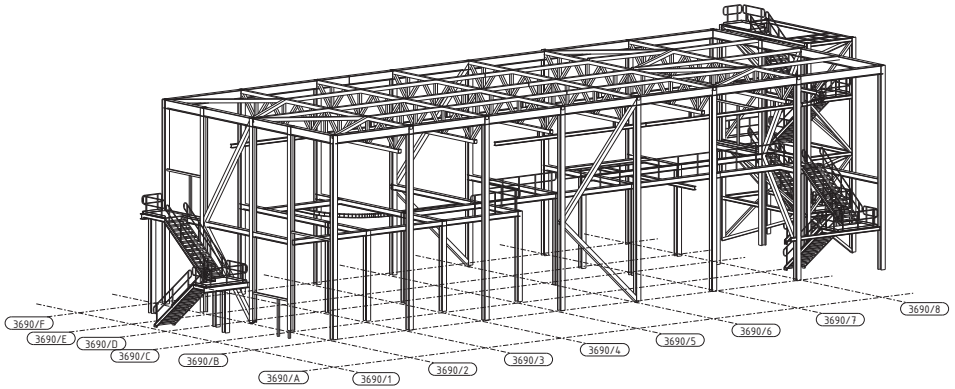
DRAWING NUMBER
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REV A01

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REV
A01

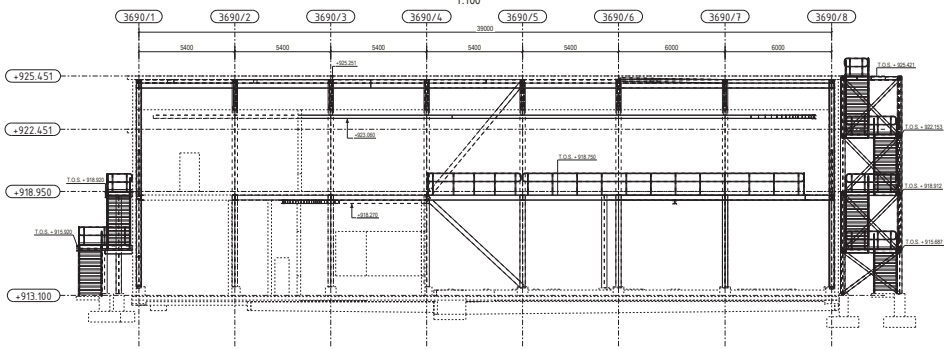
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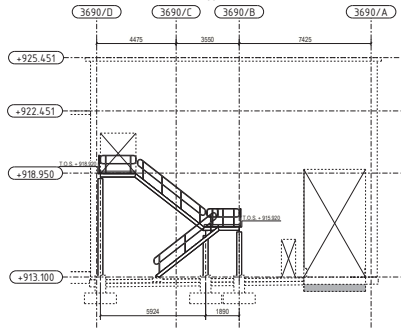
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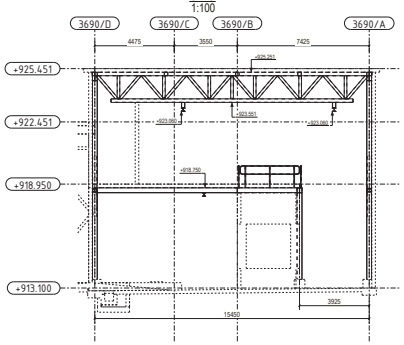
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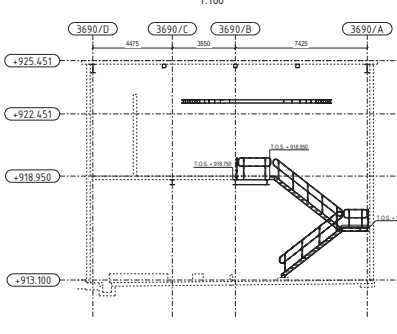
B - B
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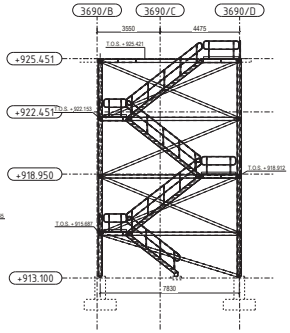
C - C
1:100



D - D
1:100



E - E
1:100



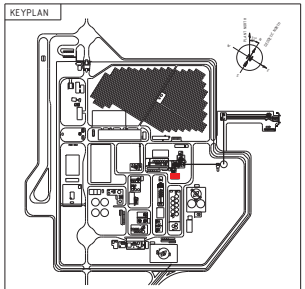
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3690 = RL(+)913.000 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.251, SECTIONS A-A, B-B AND C-C				
MD-216-3690-EG-CV-GAD-1001	PLAN AT EL +913.100	MD-216-3690-EG-CV-GAD-1002	PLAN AT EL +913.100		
MD-216-3690-EG-ME-GAD-1001	SECTION B-B	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL +913.100		
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A		
MD-216-9000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +918.950 & +925.450		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP
A01 03.01.2020 ISSUED FOR APPROVAL	Krishn	Mehul	Lodhi	Shadab	

REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP

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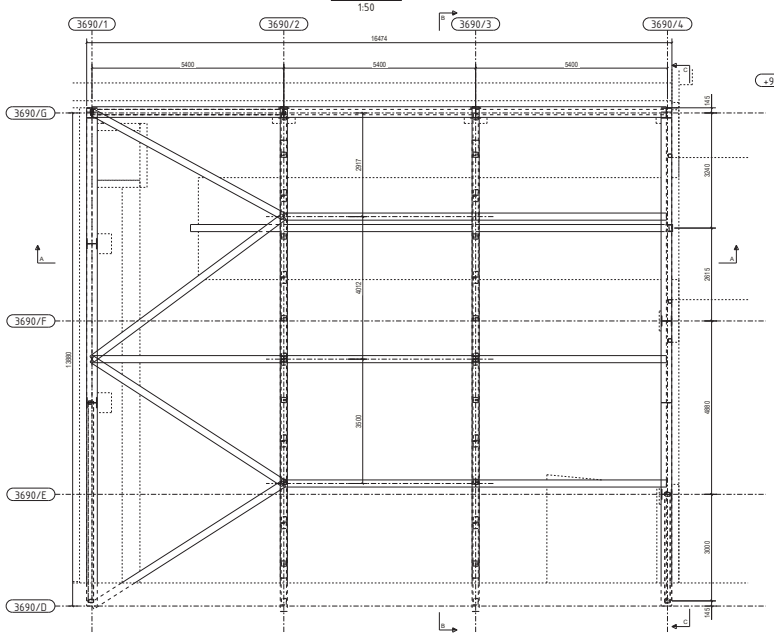
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DRAWN Nikhil	17.01.2020	
CHECKED Mehul	17.01.2020	
ENGINEER Lodhi	17.01.2020	
APPROVED Shadab	17.01.2020	
EMPLOYER		
SCALE 1:100	A0	

MANSOURAH – MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +925.251, SECTIONS A-A – E-E

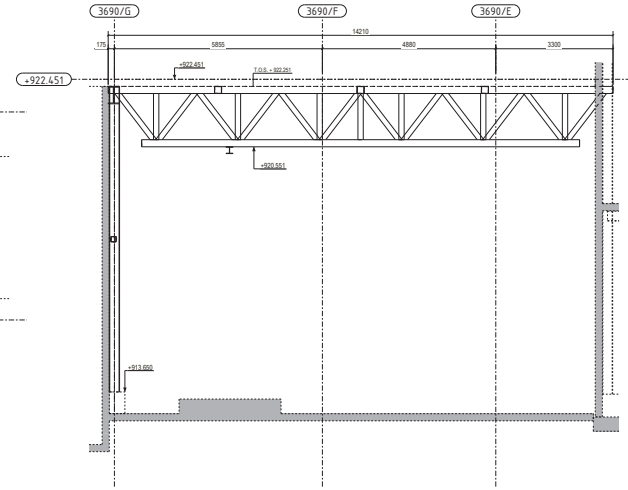
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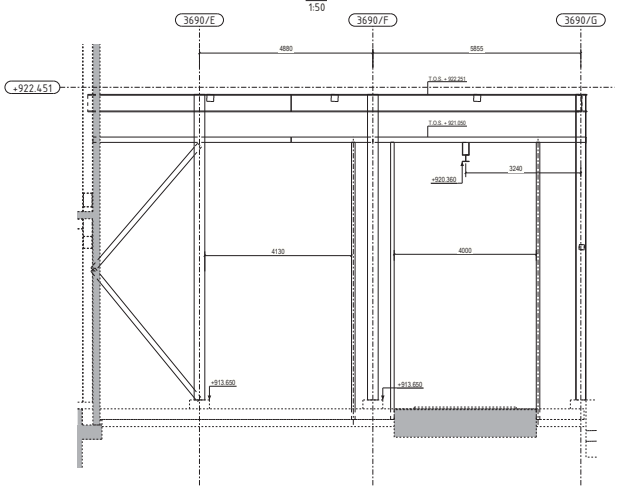
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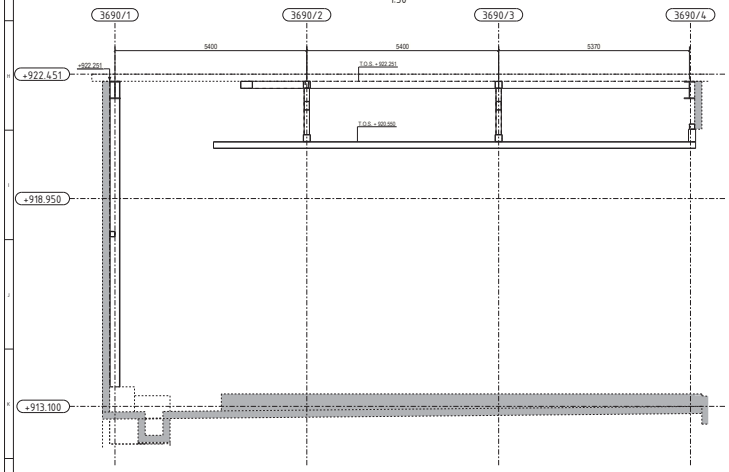
B – B
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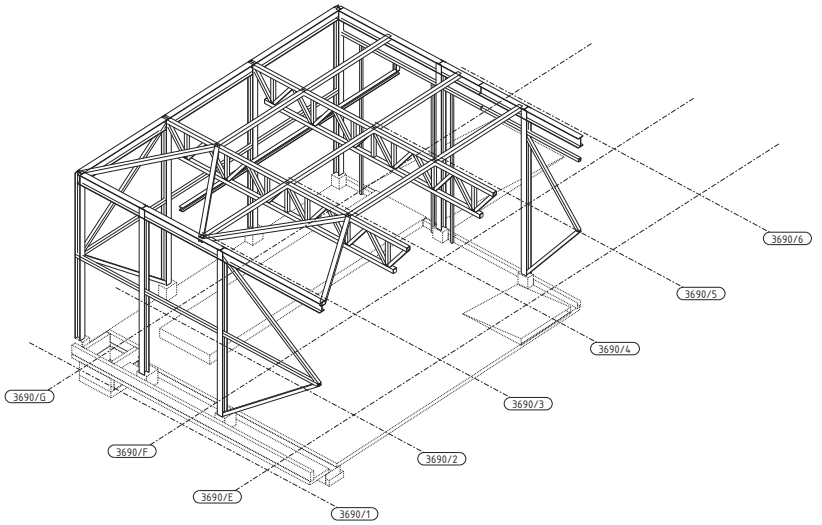
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1:50



A – A
1:50



3D
1:75



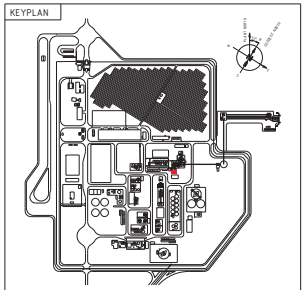
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3620 = RL(+)913.100 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3690-EG-ST-GAD-1001	PLAN AT EL. +925.251, SECTIONS A–A & C-C				
MD-216-3690-EG-CV-GAD-1001	PLAN AT EL. +910.100	MD-216-3690-EG-CV-GAD-1002	PLAN AT EL. +913.100		
MD-216-3690-EG-ME-GAD-1007	SECTION B-B	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL. +913.100		
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL. +910.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A		
MD-216-9000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL. +918.850 & +925.450		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP
A01	02.04.2020	ISSUED FOR APPROVAL	Krishn	Mehul	Sudha	Sivadas	

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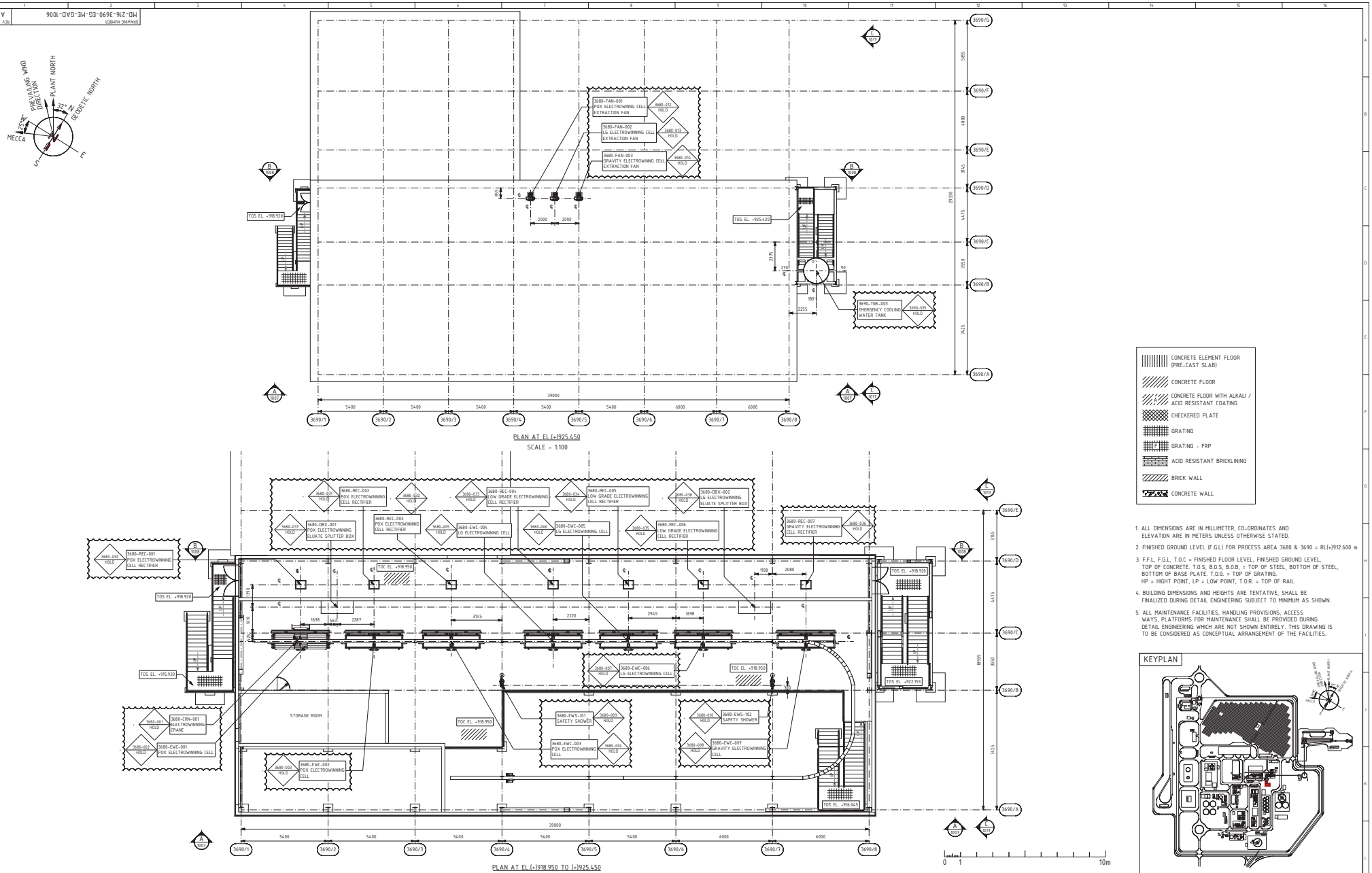
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OUTOTEC DRAWING NUMBER: OU100668225 — DOCUMENT ID — REVISION: 0

DRAWN	Ad MA	02.04.2020
CHECKED	MaLai	02.04.2020
ENGINEER	LuPra	02.04.2020
APPROVED	SivaDa	02.04.2020
EMPLOYER		
SCALE	1:50	A0

MANSOURAH – MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +922.251, SECTIONS A-A, B-B AND C-C

DRAWING NUMBER: MD-216-3690-EG-ST-GAD-1002 — REV: A01



1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L.) FOR PROCESS AREA 3680 & 3690 = RL(+)912.600 m
3. F.F.L. F.G.L. T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT, LP = LOW POINT, T.O.R. = TOP OF RAIL.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3690-EG-ME-GAD-1003	LOAD PLAN, PLAN AT EL. +913.100
MD-216-3600-EG-GM-LST-1010	HOLD LIST
MD-216-3690-EG-ME-GAD-1017	SECTION C-C
MD-216-3690-EG-ME-GAD-1008	SECTION B-B
MD-216-3690-EG-ME-GAD-1007	SECTION A-A
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL. +913.100

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP
A01	17-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP

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OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
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DRAWN	GK	17-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	17-04-2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SC	17-04-2020	GOLD ROOM, ELECTROWINNING & ICL
APPROVED	RL	17-04-2020	PLAN AT EL. +918.850 & EL. +925.450
EMPLOYER			
SCALE 1:75	A0		DRAWING NUMBER MD-216-3690-EG-ME-GAD-1006 REV A01

DRAWING NUMBER: MD-216-3690-EG-ME-GAD-1016 | REV: A01

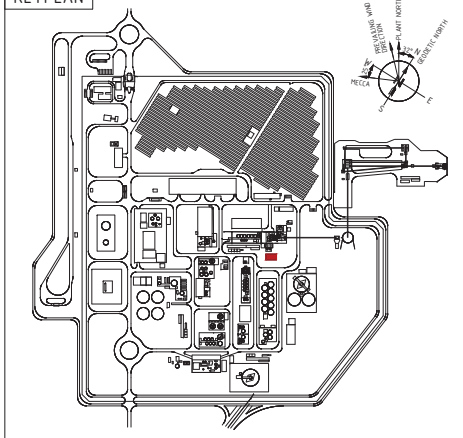
LOADING POINT	P1			P2			P3			P4.1/.../P4.4			P5.1/.../P5.2			P6			P7.1/.../P7.2			P8.1/.../P8.4			P9			P10			P11			P12			P13			P14			P15		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	- 12			- 120			- 3			- 2.5			- 5			- 12			- 5			- 2.5			- 3			- 3			- 14			- 14			- 3			- 3			- 3		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																													
3 OPERATION LOAD 1. FILLING																																													
4 OPERATION LOAD 2. PRESSURE																																													
5 OPERATION LOAD 3. TEMPERATURE																																													
6 OPERATION LOAD 4. DYNAMIC	- 18						- 5									- 18									- 5			- 5									- 5			- 5			- 5		
7 OPERATION LOAD 5. OTHERS																																													
8 TEST LOAD FILLING / EMERGENCY																																													
9 LIVE LOAD				- 425						- 10			- 20						- 20			- 10									- 90			- 90											
10 WIND ±X - DIRECTION																																													
11 WIND ±Y - DIRECTION																																													
12 EARTHQUAKE ±X - DIRECTION	-	-	0.6	-	-	21.3	-	-	-	-	-	0.5	-	-	1.0	-	-	0.6	-	-	1.0	-	-	0.5	-	-	-	-	-	-	-	4.5	-	-	4.5	-	-	-	-	-	-	-	-	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	-	-	21.3	-	-	-	-	-	0.5	-	-	1.0	-	-	0.6	-	-	1.0	-	-	0.5	-	-	-	-	-	-	-	4.5	-	-	4.5	-	-	-	-	-	-	-	-
EQUIPMENT - NO.	3620-PMP-101			3620-PKG-001			3690-PMP-001			3690-TNK-001			3690-FLT-001			3690-PMP-101			3690-FLT-002			3690-TNK-002			3690-PMP-003			3690-PMP-002			3680-TNK-003			3680-TNK-002			3680-PMP-005			3680-PMP-006			3680-PMP-004		
DESCRIPTION	INTENSIVE CYANIDATION AREA SUMP PUMP			INTENSIVE CYANIDATION VENDOR PACKAGE			POX GOLD SLUDGE FILTER PUMP			POX GOLD SLUDGE TANK			POX GOLD SLUDGE FILTER			GOLD ROOM AREA SUMP PUMP			LG GOLD SLUDGE FILTER			LG GOLD SLUDGE TANK			LG GOLD SLUDGE FILTER PUMP			LG GOLD SLUDGE FILTER PUMP			POX ELUATE RETURN TANK			POX ELUATE RETURN TANK			LG ELUATE RETURN PUMP			LG ELUATE RETURN PUMP			POX ELUATE RETURN PUMP		

LOADING POINT	P16			P17			P18			P19			P20			P21			P22			P23			P24			P25			P26			P27			P28			P29			P30		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	- 3			- 29			- 3			- 3			- 10			- 3			- 3			- 20			- 10			- 20			- 50						- 5			- 10			- 1.5		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																													
3 OPERATION LOAD 1. FILLING																																													
4 OPERATION LOAD 2. PRESSURE																																													
5 OPERATION LOAD 3. TEMPERATURE																																													
6 OPERATION LOAD 4. DYNAMIC	- 5						- 5			- 5						- 5			- 5																											
7 OPERATION LOAD 5. OTHERS																																													
8 TEST LOAD FILLING / EMERGENCY																																													
9 LIVE LOAD				- 280																		- 70			- 20						- 100														
10 WIND ±X - DIRECTION															0.5																														
11 WIND ±Y - DIRECTION																		0.5																											
12 EARTHQUAKE ±X - DIRECTION	-	-	-	-	14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3.5	-	-	1	-	-	1.0	-	-	5	-	-	0.5	-	-	-	-	-	-	-	-	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	-	-	14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3.5	-	-	1	-	-	1.0	-	-	5	-	-	0.5	-	-	-	-	-	-	-	-	-
EQUIPMENT - NO.	3680-PMP-003			3680-TNK-001			3680-PMP-001			3680-PMP-002			3690-CLT-001			3690-PMP-004			3690-PMP-005			3690-CRN-001			3690-FNC-002			-			3690-FNC-001			-			3690-MSC-006			3690-FLT-003			3680-REC-001		
DESCRIPTION	POX ELUATE RETURN PUMP			GRAVITY ELECTROWINNING LEACHATE TANK			LEACHATE PUMP			LEACHATE PUMP			FURNACE COOLING TOWER			FURNACE COOLING WATER PUMP			FURNACE COOLING WATER PUMP			GOLD ROOM JIB CRANE			SMELTING FURNACE			SMELTING FURNACE INVERTER			CALCINING OVEN TRAY RACK			FLUX MIXER			SMELT FURNACE DUST COLLECTOR			POX ELECTROWINNING CELL RECTIFIER		

LOADING POINT	P31			P32			P33			P34			P35			P36			P37.1/.../P37.4			P38.1/.../P38.4			P39.1/.../P39.6			P40.1/.../P40.6			P41.1/.../P41.6		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	- 1.5			- 1.5			- 1.5			- 1.5			- 1.5			- 1.5			- 1.25			- 0.25			- 3.3			- 3.3			- 3.3		
2 DEAD LOAD (SECONDARY EQUIPMENT)																																		
3 OPERATION LOAD 1. FILLING																																		
4 OPERATION LOAD 2. PRESSURE																																		
5 OPERATION LOAD 3. TEMPERATURE																																		
6 OPERATION LOAD 4. DYNAMIC																																		
7 OPERATION LOAD 5. OTHERS																																		
8 TEST LOAD FILLING / EMERGENCY																																		
9 LIVE LOAD																			- 7.5			- 7.5			- 5			- 5			- 5		
10 WIND ±X - DIRECTION																																		
11 WIND ±Y - DIRECTION																																		
12 EARTHQUAKE ±X - DIRECTION	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.4	-	-	0.4	-	-	0.25	-	-	0.25	-	-	0.25	-	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.4	-	-	0.4	-	-	0.25	-	-	0.25	-	-	0.25	-	-	0.25
EQUIPMENT - NO.	3680-REC-002			3680-REC-003			3680-REC-004			3680-REC-005			3680-REC-006			3680-REC-007			3680-DBX-001			3680-DBX-002			3680-EWC-001			3680-EWC-002			3680-EWC-003		
DESCRIPTION	POX ELECTROWINNING CELL RECTIFIER			POX ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			GRAVITY ELECTROWINNING CELL RECTIFIER			POX ELECTROWINNING ELUATE SPLITTER BOX			LG ELECTROWINNING ELUATE SPLITTER BOX			POX ELECTROWINNING CELL			POX ELECTROWINNING CELL			POX ELECTROWINNING CELL		

LOADING POINT	P42.1/.../P42.6			P43.1/.../P43.6			P44.1/.../P44.6			P45.1/.../P45.6			P46			P47			P48			P49		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	- 3.3			- 3.3			- 3.3			- 3.3			- 3			- 3			- 3			- 12		
2 DEAD LOAD (SECONDARY EQUIPMENT)																								
3 OPERATION LOAD 1. FILLING																								
4 OPERATION LOAD 2. PRESSURE																								
5 OPERATION LOAD 3. TEMPERATURE																								
6 OPERATION LOAD 4. DYNAMIC																								
7 OPERATION LOAD 5. OTHERS																								
8 TEST LOAD FILLING / EMERGENCY																								
9 LIVE LOAD	- 5			- 5			- 5			- 5												- 80		
10 WIND ±X - DIRECTION															0.2			0.2			0.2			
11 WIND ±Y - DIRECTION																								
12 EARTHQUAKE ±X - DIRECTION	-	-	0.25	-	-	0.25	-	-	0.25	-	-	0.25	-	-	-	-	-	-	-	-	-	-	4	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	0.25	-	-	0.25	-	-	0.25	-	-	0.25	-	-	-	-	-	0.2	-	-	0.2	-	-
EQUIPMENT - NO.	3680-EWC-004			3680-EWC-005			3680-EWC-006			3680-EWC-007			3680-FAN-001			3680-FAN-002			3680-FAN-003			3690-TNK-003		
DESCRIPTION	LG ELECTROWINNING CELL			LG ELECTROWINNING CELL			LG ELECTROWINNING CELL			GRAVITY ELECTROWINNING CELL			POX ELECTROWINNING CELL EXTRACTION FAN			LG ELECTROWINNING CELL EXTRACTION FAN			GRAVITY ELECTROWINNING CELL EXTRACTION FAN			EMERGENCY COOLING WATER TANK		

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3690-EG-ME-GAD-1004	LOAD PLAN, PLAN AT EL. +918.950 & EL. +925.450								
MD-216-3690-EG-ME-GAD-1003	LOAD PLAN, PLAN AT EL. +913.100	A01	08-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_D_02003_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707 __GAU01 __EE14 __MLD14 __36903	DOCUMENT ID OU100680654	REVISION 0
DRAWN	GK	08-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	08-04-2020	LOAD PLAN
ENGINEER	SC	08-04-2020	GOLD ROOM, ELECTROWINNING & ICL
APPROVED	RL	08-04-2020	LOAD TABLE
EMPLOYER			
SCALE	-	A1	DRAWING NUMBER MD-216-3690-EG-ME-GAD-1016 REV A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	C01	D01	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

ATTACHMENT 2 : STAAD INPUT FILE OF GOLD ROOM

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 12-May-20
JOB NAME MMGP/GOLD ROOM
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SNM
CHECKER NAME SJM
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 913.45 0; 2 3.2 913.45 0; 3 0 913.45 8.025; 4 3.2 913.45 8.025;
5 0 918.45 0; 6 3.2 918.45 0; 7 0 918.45 8.025; 8 3.2 918.45 8.025;
9 0 924.95 0; 10 3.2 924.95 0; 11 0 924.95 8.025; 12 3.2 924.95 8.025;
13 0 915.25 8.025; 14 3.2 915.25 8.025; 15 0 921.7 8.025; 16 3.2 921.7 8.025;
17 0 915.25 0; 18 3.2 915.25 0; 19 0 921.7 0; 20 3.2 921.7 0; 21 0 918.45 2;
22 0 921.7 2; 24 3.2 924.95 4.0125; 25 0 924.95 4.0125; 26 3.2 921.7 4.0125;
27 3.2 915.25 4.0125; 28 0 915.25 4.0125; 29 0 918.45 4.0125;
30 0 921.7 4.0125; 31 3.2 924.95 5.825; 32 0 924.95 5.825; 33 3.2 924.95 6.925;
34 0 924.95 6.925; 35 0.8 924.95 8.025; 36 1.6 924.95 8.025;
37 2.4 924.95 8.025; 38 0.8 924.95 5.825; 39 1.6 924.95 5.825;
40 2.4 924.95 5.825; 41 0.8 924.95 6.925; 42 1.6 924.95 6.925;
43 2.4 924.95 6.925; 44 3.2 918.45 4.0125;
MEMBER INCIDENCES
1 1 17; 2 2 18; 3 3 13; 4 4 14; 5 5 6; 6 6 44; 7 5 21; 8 7 8; 9 5 19; 10 6 20;
11 7 15; 12 8 16; 13 9 10; 14 10 24; 15 9 25; 16 11 35; 17 13 7; 18 14 8;
19 13 14; 20 15 11; 21 16 12; 22 15 16; 23 17 5; 24 18 6; 25 19 9; 26 20 10;
27 19 20; 28 17 18; 29 20 26; 30 19 22; 31 18 27; 32 17 28; 33 1 18; 34 18 5;
35 5 20; 36 20 9; 37 3 14; 38 14 7; 39 7 16; 40 16 11; 41 21 29; 42 22 30;
43 21 22; 45 24 31; 46 25 32; 47 26 16; 48 27 14; 49 28 13; 50 1 28; 51 28 3;
52 29 7; 53 30 15; 70 21 30; 71 30 7; 72 17 29; 73 29 13; 74 19 25; 75 25 15;
76 4 27; 77 27 6; 78 27 8; 79 6 26; 80 26 8; 81 20 24; 82 24 16; 83 31 33;
84 32 34; 85 32 38; 86 33 12; 87 34 11; 88 34 41; 89 35 36; 90 36 37; 91 37 12;
92 38 39; 93 39 40; 94 40 31; 95 41 42; 96 42 43; 97 43 33; 98 40 43; 99 43 37;
100 39 42; 101 42 36; 102 38 41; 103 41 35; 104 44 8; 105 27 44;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 4 9 TO 12 17 18 20 21 23 TO 26 TABLE ST IPE240
```

```
5 19 22 43 TABLE ST IPE240
8 27 28 33 36 37 40 105 TABLE SD L65X65X6 SP 0.008
6 29 31 32 47 TO 49 104 TABLE SD L90X90X8 SP 0.008
34 35 38 39 50 51 70 TO 82 TABLE SD L75X75X8 SP 0.008
7 13 TO 16 30 41 42 45 46 52 53 83 TO 103 TABLE ST HE220A
CONSTANTS
BETA 90 MEMB 1 TO 4 9 TO 12 17 18 20 21 23 TO 26 43
MATERIAL STEEL ALL
SUPPORTS
1 TO 4 PINNED
MEMBER TRUSS
33 TO 40 50 51 70 TO 82 105
MEMBER RELEASE
5 6 8 13 TO 16 19 22 28 29 31 32 43 85 88 98 TO 103 START MZ
5 8 13 19 22 27 28 43 47 TO 49 52 53 86 87 91 94 97 TO 104 END MZ
**********************
DEFINE IBC 2003
SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.35 PZ 0.35
SELFWEIGHT 1
MEMBER WEIGHT
5 13 19 22 CON 2.2 0.3
5 13 19 22 CON 2.2 1.5
5 19 22 CON 2.2 1.7
5 19 22 CON 2.2 2.9
***************
LOAD 11 EARTHQUAKE LOAD (+X)
IBC LOAD X 1
LOAD 12 EARTHQUAKE LOAD (+Z)
IBC LOAD Z 1
***************
LOAD 1 DEAD LOAD (DL)
SELFWEIGHT Y -1
MEMBER LOAD
5 13 19 22 CON GY -2.2 0.3
5 13 19 22 CON GY -2.2 1.5
5 19 22 CON GY -2.2 1.7
5 19 22 CON GY -2.2 2.9
**********************
LOAD 2 LIVE LOAD (LL)
MEMBER LOAD
5 13 19 22 CON GY -9.6 0.3
5 13 19 22 CON GY -9.6 1.5
5 19 22 CON GY -9.6 1.7
5 19 22 CON GY -9.6 2.9
**********************
LOAD 3 SUPERIMPOSED LOAD (SDL1)
JOINT LOAD
11 12 31 TO 43 FX 0.27 FY -6.13 FZ 0.27
**********************
LOAD 4 SUPERIMPOSED LOAD (SDL2)
```

```
JOINT LOAD
11 12 31 TO 43 FX -0.27 FY -6.13 FZ -0.27
**********************
LOAD 5 WIND LOAD +X (WL+X)
MEMBER LOAD
1 TO 4 9 TO 12 17 18 20 21 23 TO 26 UNI GX 1.49
JOINT LOAD
11 12 31 TO 43 FX 0.7
**********************
LOAD 6 WIND LOAD -X (WL-X)
MEMBER LOAD
1 TO 4 9 TO 12 17 18 20 21 23 TO 26 UNI GX -1.49
JOINT LOAD
11 12 31 TO 43 FX -0.7
**********************
LOAD 7 WIND LOAD +Z (WL+Z)
MEMBER LOAD
1 TO 4 9 TO 12 17 18 20 21 23 TO 26 UNI GZ 0.53
JOINT LOAD
11 12 31 TO 43 FZ 0.7
**********************
LOAD 8 WIND LOAD -Z (WL-Z)
MEMBER LOAD
1 TO 4 9 TO 12 17 18 20 21 23 TO 26 UNI GZ -0.53
JOINT LOAD
11 12 31 TO 43 FZ -0.7
**********************
LOAD 9 TEMPERATURE LOAD 1 (TL1)
TEMPERATURE LOAD
1 TO 43 45 TO 53 70 TO 105 TEMP 25
**********************
LOAD 10 TEMPERATURE LOAD 2 (TL2)
TEMPERATURE LOAD
1 TO 43 45 TO 53 70 TO 105 TEMP -25
**********************
LOAD 101 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 102 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 103 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 104 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 105 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
```

```
LOAD 106 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 107 DL + SDL1 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 108 DL + SDL1 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 109 DL + SDL1 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 110 DL + SDL1 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 111 DL + SDL2 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 112 DL + SDL2 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 113 DL + SDL2 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 114 DL + SDL2 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 115 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.75
LOAD 116 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.75
LOAD 117 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.75
LOAD 118 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.75
LOAD 119 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.75
LOAD 120 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.75
LOAD 121 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.75
LOAD 122 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
```

```
1 1.0 4 1.0 2 0.75 12 -0.75
LOAD 123 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 124 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 125 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 126 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 127 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 128 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 129 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 130 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 131 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 132 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 133 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 134 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 135 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 136 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 137 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 0.7
LOAD 138 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
```

```
*Load Combination for Structural design
LOAD 201 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 202 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 203 1.2 (DL + SDL1 + TL1) + 1.6 LL
REPEAT LOAD
1 1.2 3 1.2 9 1.2 2 1.6
LOAD 204 1.2 (DL + SDL1 + TL2) + 1.6 LL
REPEAT LOAD
1 1.2 3 1.2 10 1.2 2 1.6
LOAD 205 1.2 (DL + SDL2 + TL1) + 1.6 LL
REPEAT LOAD
1 1.2 4 1.2 9 1.2 2 1.6
LOAD 206 1.2 (DL + SDL2 + TL2) + 1.6 LL
REPEAT LOAD
1 1.2 4 1.2 10 1.2 2 1.6
LOAD 207 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 208 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 209 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 210 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 211 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 212 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 213 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 214 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 215 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 216 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 217 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
```

```
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 218 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 219 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 220 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 221 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 222 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 223 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 224 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 225 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 226 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 227 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 228 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 229 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 230 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 231 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 232 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 233 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
```

```
LOAD 234 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 235 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 236 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 237 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 238 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
PERFORM ANALYSIS
STEEL TAKE OFF ALL
LOAD LIST 201 TO 238
PARAMETER 1
CODE LRFD
BEAM 1 ALL
NSF 0.9 ALL
LY 7.7 MEMB 6 7 29 TO 32 41 42 47 TO 49 52 53 104
LZ 3.8 MEMB 45 46 83 84 86 87
LZ 3.2 MEMB 16 85 88 TO 97
KZ 0.85 MEMB 33 TO 40 50 51 70 TO 82 105
KY 0.85 MEMB 33 TO 40 50 51 70 TO 82 105
CHECK CODE ALL
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	C01	D01	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

ATTACHMENT 3 : COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: MAADEN-TRANSMIT-002660, 24 June 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 24 June 2020	Comment by	EPC Response DATE: 29 June 2020	Status	Remarks
			General Comments (cc)				
MD-216-3690-EG-CV-CAL-1372-C01	3690	1	SBC 304	Bechtel / Ma'aden	Rev year of SBC 304 has been change to 2018 from 2007	Closed	